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Platform Economics and Online Marketplaces

Dr. Chaoran Liu

Lecture 3
Platform Pricing

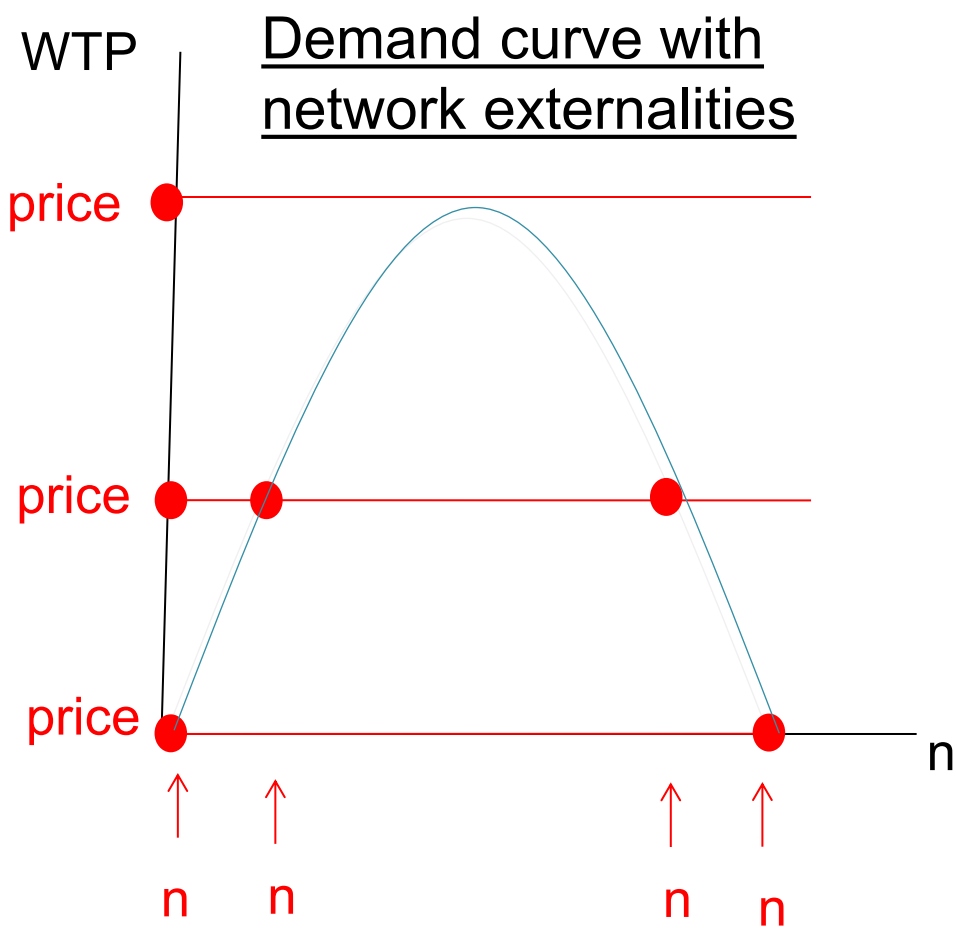
Peking University HSBC Business School

Recap

Demand with Network Effects

Critical Mass

Demand Curve with Network Effect



- If the price is higher than the highest WTP, n = ?

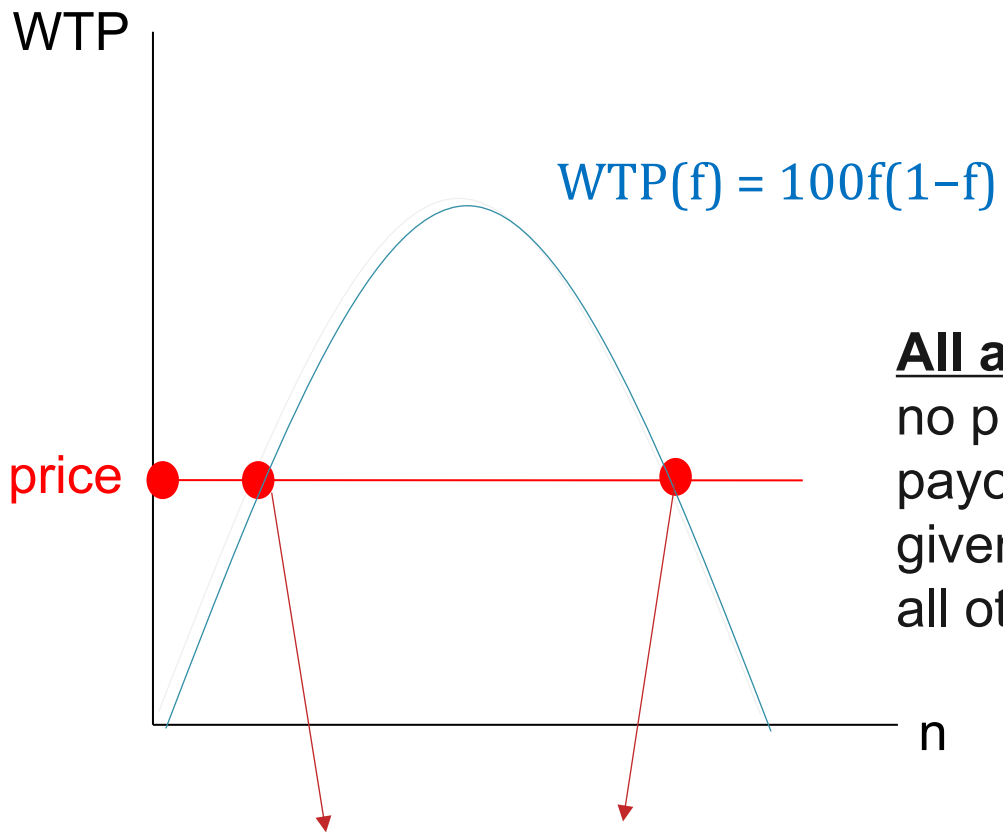
  n = 0 (or very small)

- If the price is zero, n = ?

  any value along the horizontal axis

- If the price is between zero and the highest WTP, n = ?

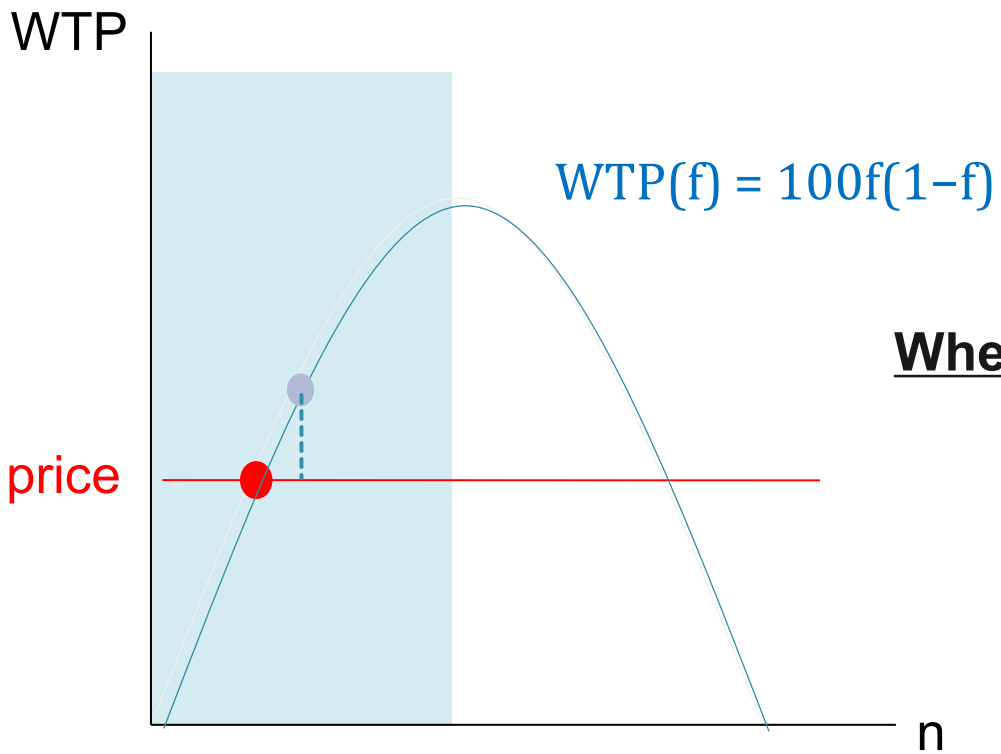
Demand Curve with Network Effect



<u>All are Nash Equilibria:</u>
no player can increase their payoff by unilaterally deviating, given the strategies chosen by all other players (f fixed)

WTP(f*) = p when marginal customer is indifferent

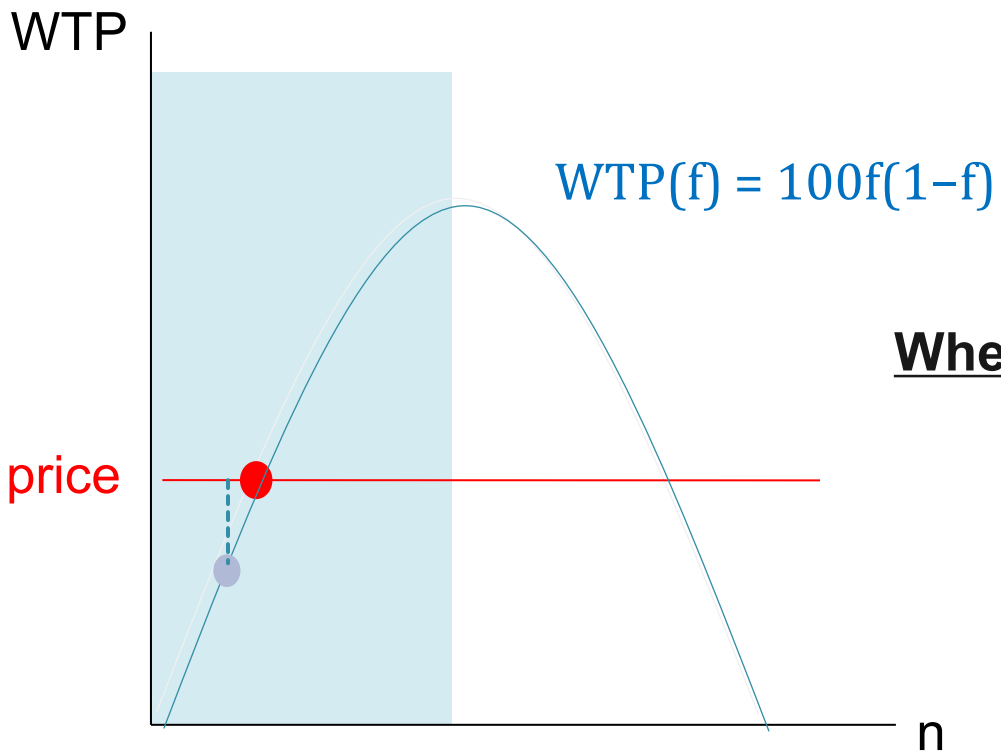
Demand Curve with Network Effect



When f deviates + →

- WTP > p
- F continues to grow (WTP >p)
- Not going to self correct

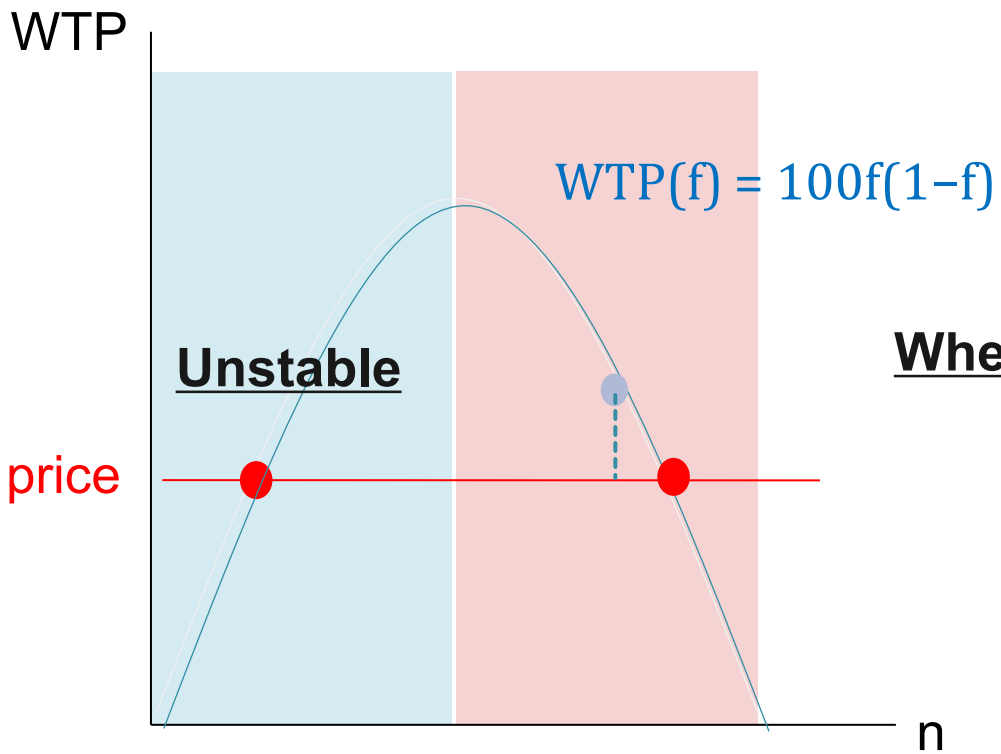
Demand Curve with Network Effect



When f deviates - →

- WTP < p
- F continues to decrease (WTP <p)
- Not going to self correct

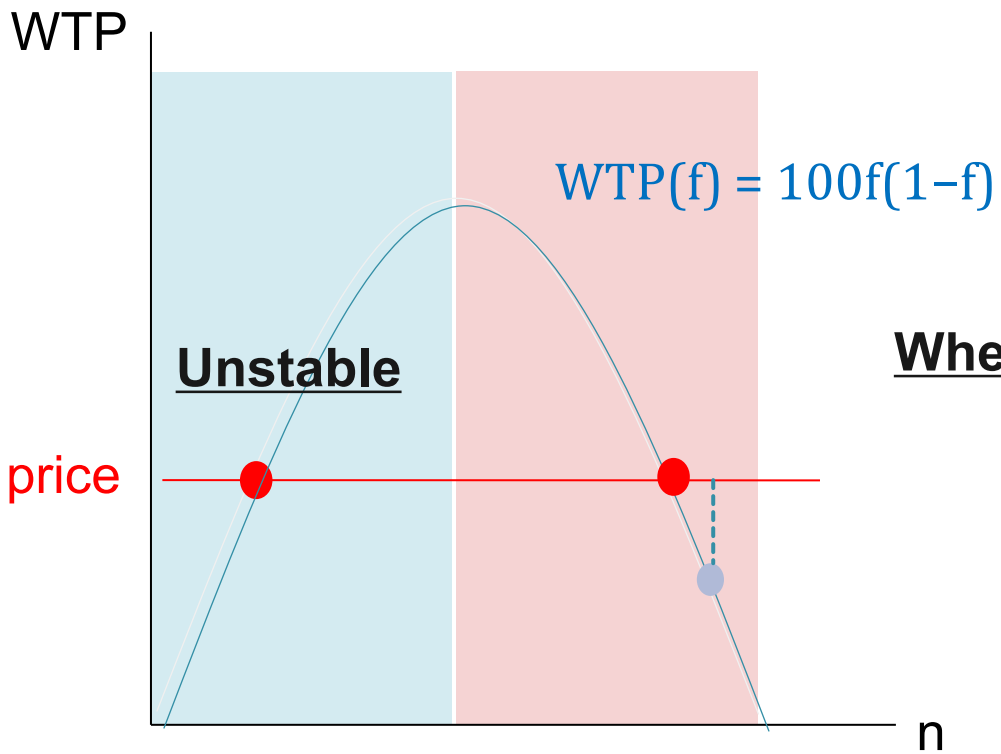
Demand Curve with Network Effect



When f deviates - →

- WTP > p
- F continues to grow (WTP >p)
- Self correct back

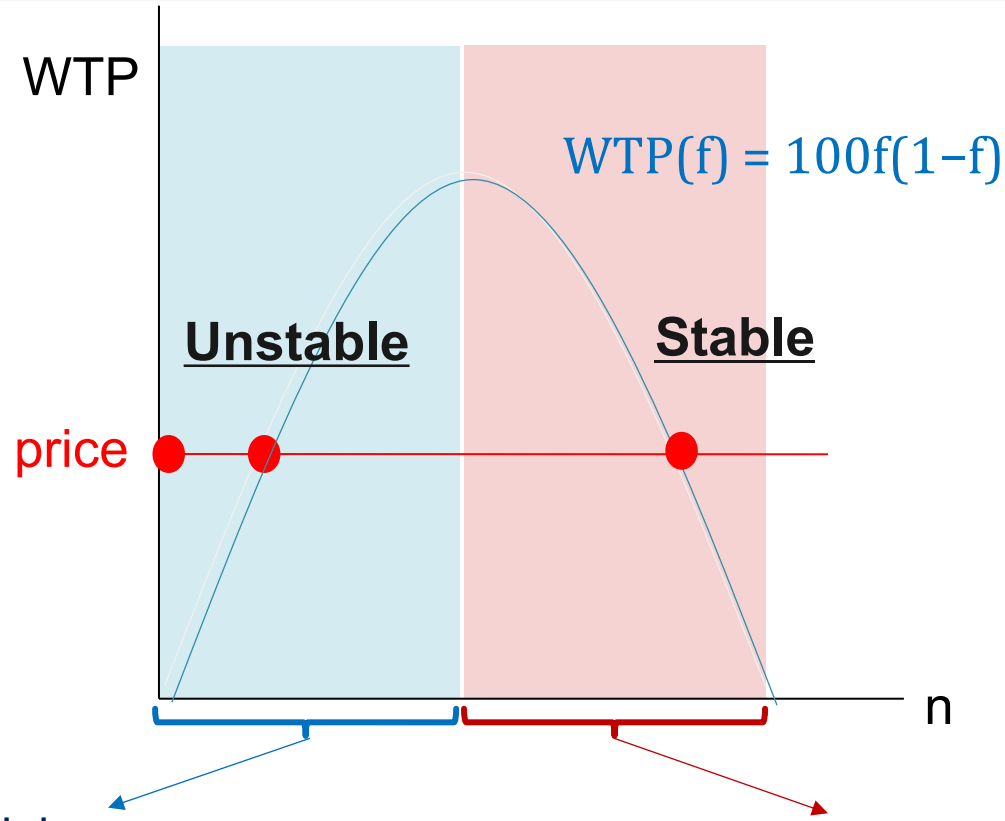
Demand Curve with Network Effect



When f deviates + →

- WTP < p
- F continues to shrink (WTP <p)
- Self correct back

Demand Curve with Network Effect



As more people join:
The product becomes more useful.
More consumers are willing to pay.

→ Self-reinforcing demand
→ Critical mass

The WTP does not increase forever

- Consumers are heterogeneous.
- Start attracting lower-intrinsic-value consumers

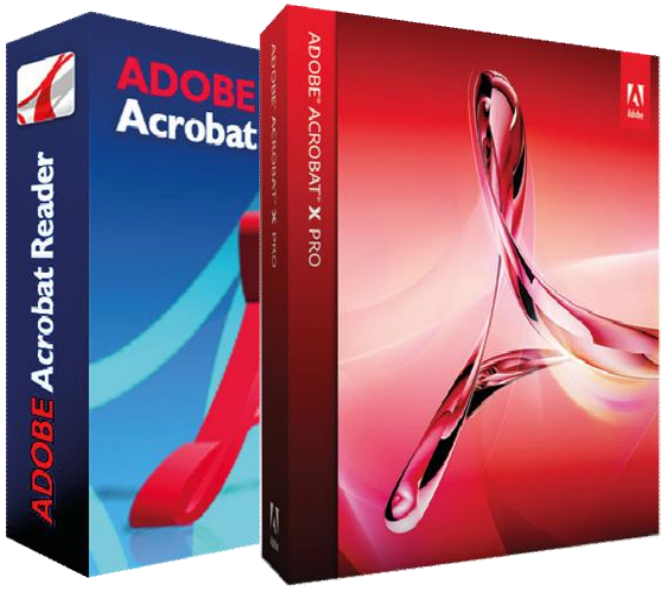
→ There is a limit to expansion (saturation level)
→ Late adopters are harder to convince

Roadmap

- Some puzzles
- Review of standard monopoly pricing
- Platform pricing
  - Transaction fee case
  - Registration fee case
  - Divide-and-conquer
- Platform pricing practice

Adobe Acrobat

- The Story of Acrobat Reader/Writer
  - Adobe Acrobat PDF reader is free while the Adobe Acrobat writer is not.
  - Why?



PC vs. Video Game Console

- End users favour platforms that offer a wide variety of complements. Developers favor platforms with more end users.

Gamer

Developer

In video games, **<u>end users are subsidized</u>**. Platform providers like Sony PlayStation and Microsoft Xbox price consoles at or below cost. Game developers pay a royalty to console manufacturers as much as 20% of a game retail price.

PC vs. Video Game Console

- End users favour platforms that offer a wide variety of complements. Developers favour platforms with more end users.

User

Developer

In the PC market, **<u>end users are the money side</u>**, paying well above cost. Application developers, on the other hand, pay no royalties and receive free software development kits from the OS vendors.

PC vs. Video Game Console

- Different pricing strategies
  - Why?

More Puzzles

- A night club allows women to enter free. But it charges men entrance fee.
- Youtube allows us to visit and see its content for free. But advertisers need to pay.

Asymmetric Prices Structure

Platform	profit making group	subsidized group
Google	advertisers	searchers
Yahoo!	advertisers	viewers
PayPal	merchants	consumers
eBay	auctioneers	bidders
Wall Street Journal	readers	columnists

Asymmetric Prices Structure

- It seems to be related to the network effect. But how? Any other factors?

Roadmap

- Some puzzles
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Standard Monopoly Pricing

- A monopoly firm faces consumer demand $D(p)$ where $D'(p) < 0$.
- Constant marginal cost is $c > 0$.
- The monopoly maximizes profits by choosing p.

$$\max_{p} \ (p - c)D(p)$$

- The F.O.C. yields

$$(p^* - c)D'(p^*) + D(p^*) = 0,$$

where p^* denotes the optimal price.

Standard Monopoly Pricing

- By rearranging the F.O.C., we derive a formula for optimal pricing:

$$p^* - c = -\frac{D(p)}{D'(p)}.$$

Note that the L.H.S. is the monopolist's profit margin.

- The equation can be written in a more familiar fashion:

$$\underbrace{\frac{p^* - c}{p^*}}_{\text{Lerner index}} = \frac{1}{\epsilon}$$

where $\epsilon = -p^* D'(p^*)/D(p^*)$, is the price elasticity of demand.

Standard Monopoly Pricing

- $\epsilon = -p^* D'(p^*)/D(p^*)$

$$\epsilon = -\frac{\%\ change\ in\ quantity}{\%\ change\ in\ price} = \frac{dQ/Q}{dp/p}$$

$$= -\frac{D'(p^*)}{D(p^*)} p^*$$

Standard Monopoly Pricing

$$\underbrace{\frac{p^* - c}{p^*}}_{\text{Lerner index}} = \frac{1}{\epsilon}$$

→ How much a firm can price above marginal cost

- Perfect competition: P=C → Lerner = 0
- Monopoly: Lerner > 0

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Basics in Pricing

- Two types of fees
  - Transaction fees
    - Paid per effective transaction
  - Membership fees:
    - Paid once or renewed on a regular basis
- Who would platform charge?
  - Examples:
    - Uber and Lyft charge drivers a commission
    - Airbnb charges both guest and hosts commission

Two-sided Monopoly: Transaction Fees

- Assume no cost of joining the platform
- Two distinct groups, buyers B and sellers S, have to transact with each other via a monopoly platform.
  - buyers' (gross) surplus per transaction: v_B is distributed over $[0\,,1]$.
  - sellers' (gross) surplus per transaction: v_S is distributed over $[0\,,1]$.

Two-sided Monopoly: Transaction Fees

- Assume no cost of joining the platform
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  - buyers' (gross) surplus per transaction: v_B is distributed over $[0\,,1]$.
  - sellers' (gross) surplus per transaction: v_S is distributed over $[0\,,1]$.
- Cross-group network externality:
  - buyers' (gross) total surplus: $v_B D_S$ where D_S is the number of sellers who use the platform.
  - sellers' (gross) total surplus: $v_S D_B$ where D_B is the number of buyers who use the platform.

Two-sided Monopoly: Transaction Fees

- The monopoly platform charges fees per transaction:
  - t_B to buyers
  - t_S to sellers
- The platform incurs cost $c > 0$ per transaction.
- Participation: *Size of the other network does not affect platform choice*
  - a buyer expects net surplus $(v_B - t_B)D_S$ from using the platform. He uses the platform iff $v_B - t_B > 0$.
  - a seller expects net surplus $(v_S - t_S)D_B$ from using the platform. He uses the platform iff $v_S - t_S > 0$.
- Demand:
  - demand from buyer side: $D_B(t_B) = \Pr(v_B - t_B > 0)$.
  - demand from seller side: $D_S(t_S) = \Pr(v_S - t_S > 0)$.

Two-sided Monopoly: Transaction Fees

- Total transaction volume: $D_B(t_B)D_S(t_S)$.
- The monopoly **platform** maximizes profits by choosing t_B and t_S :

$$\max_{t_B,t_S} (t_B + t_S - c)D_B(t_B)\, D_S(t_S)$$

- Then the F.O.C.s are

$$(t_i^* + t_j^* - \ c)D_i' + \ D_i \ = \ 0, \forall i \ = \ B, S \ and \ j \neq i.$$

- Sample steps from Buyer side:
  - F.O.C to t_B

$$(t_B + t_S - \ c)D_B' D_s + \ D_B D_s = \ 0$$
$$(t_B + t_S - \ c)D_B' + \ D_B = \ 0$$

Two-sided Monopoly: Transaction Fees

- We can again write the optimal transaction fees in term of price elasticity

$$t_i^* + t_j^* - c = \frac{t_i^*}{\epsilon_i}, \forall i = B, S \text{ and } j \neq i$$

where $\epsilon_i = -t_i^* D_i'(t_i^*)/D_i(t_i^*)$ is the price elasticity of demand of side i.

- Sample steps from Buyer side:

$$(\mathrm{t_B} + \mathrm{t_S} - \mathrm{c}) = -\frac{\mathrm{D_B}}{\mathrm{D_B'}} = -\frac{\mathrm{t_B}}{\frac{\mathrm{D_B'}}{\mathrm{D_B}} * \mathrm{t_B}}$$

$$= \frac{t_B}{\epsilon_B}$$

Two-sided Monopoly: Transaction Fees

- This leads to

$$(t_B + t_S - c) = \boxed{\frac{t_B}{\epsilon_B} = \frac{t_S}{\epsilon_S}}$$

$$\boxed{\frac{t_B}{t_S} = \frac{\epsilon_B}{\epsilon_S}}$$

Under the profit-maximizing price structure, the ratio of transaction fees is equal to the ratio of demand elasticities

Two-sided Monopoly: Transaction Fees

- **Counterintuitive**: if buyers are insensitive to **transaction fee** changes (low elasticity), they will face low transaction fee

Roadmap

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Two-sided Monopoly: Membership Fees

- Now consider, instead of charging fee based on transaction volumes, the monopoly platform charges one-time membership (registration) fee.
  - r_B to buyers
  - r_S to sellers.
- For simplicity, we assume
  - the total value of each transaction is 1.
  - a buyer's surplus share is α_B while a seller's surplus share is α_S.
  - $\alpha_B + \alpha_S = 1$

Two-sided Monopoly: Membership Fees

- Utilities:
  - buyers: $u_B = \alpha_B D_S - r_B$.
  - sellers: $u_s = \alpha_S D_B - r_S$.
  - Note that the disutility caused by r_B is no longer $r_B D_S$, the disutility caused by r_s is no longer $r_s D_B$. (membership fee)
  - u_B is affected by r_B and r_s. (Why?)

- D_B is a function of u_B, i.e., $D_B(u_B)$. So D_B is also a function of r_B, r_S, i.e., $D_B(r_B, r_S)$.

Two-sided Monopoly: Membership Fees

- A potential problem: equilibrium multiplicity.
  - if one side expects zero participation from the other side, then they will not participate either.
  - this may require to use negative fee to subsidize one side.
  - for now we focus on equilibria with positive participation.

- Assume no transaction cost, but the platform incurs cost of serving each participant: c_B and c_S.
- **<u>Platform's problem</u>** is then

$$\max_{r_B, r_S} \quad (r_B - c_B)\, D_B(r_B, r_S) + (r_S - c_S)\, D_S(r_B, r_S)$$

Two-sided Monopoly: Membership Fees

- One simple way to solve the maximization is to let the platform directly choose the utilities it offers to the two sides
  - $u_B = \alpha_B D_S - r_B$ and $u_S = \alpha_S D_B - r_S$.
  - the demands are then $D_B(u_B)$ and $D_S(u_S)$.
- The platform's maximization can be re-formulated to

$$\max_{u_S, u_B} \quad (\alpha_B D_S(u_S) - u_B - c_B)\, D_B(u_B) + (\alpha_S D_B(u_B) - u_S - c_S)\, D_S(u_S)$$

- The F.O.C.s yield

$$\left(\alpha_i D_j - u_i - c_i\right) D_i' - D_i + \alpha_j D_j D_i' = 0,$$
$$\forall i = B, S \text{ and } j \neq i$$

Two-sided Monopoly: Membership Fees

- Rearranging the F.O.C.s, the optimal registration fees are determined by

$$r_i^* - c_i + \underbrace{\alpha_j \cdot D_j}_{\text{network effect}} = \frac{D_i}{D_i'},$$

or

$$\frac{r_i^* - (c_i - \alpha_j D_j)}{r_i^*} = \frac{1}{\epsilon_i}$$

Two-sided Monopoly: Membership Fees

$$\frac{r_i^* - (c_i - \alpha_j D_j)}{r_i^*} = \frac{1}{\epsilon_i}$$

- To maximize profits on a platform that charges membership fees
  - a lower membership fee is charged to the group of users that exerts the larger network benefit on the other
  - It may subsidize this group so as to generate a higher volume of trade and extract higher profits from the other group of users
- Magazine: subsidize the readers to make the magazine more attractive for advertisers

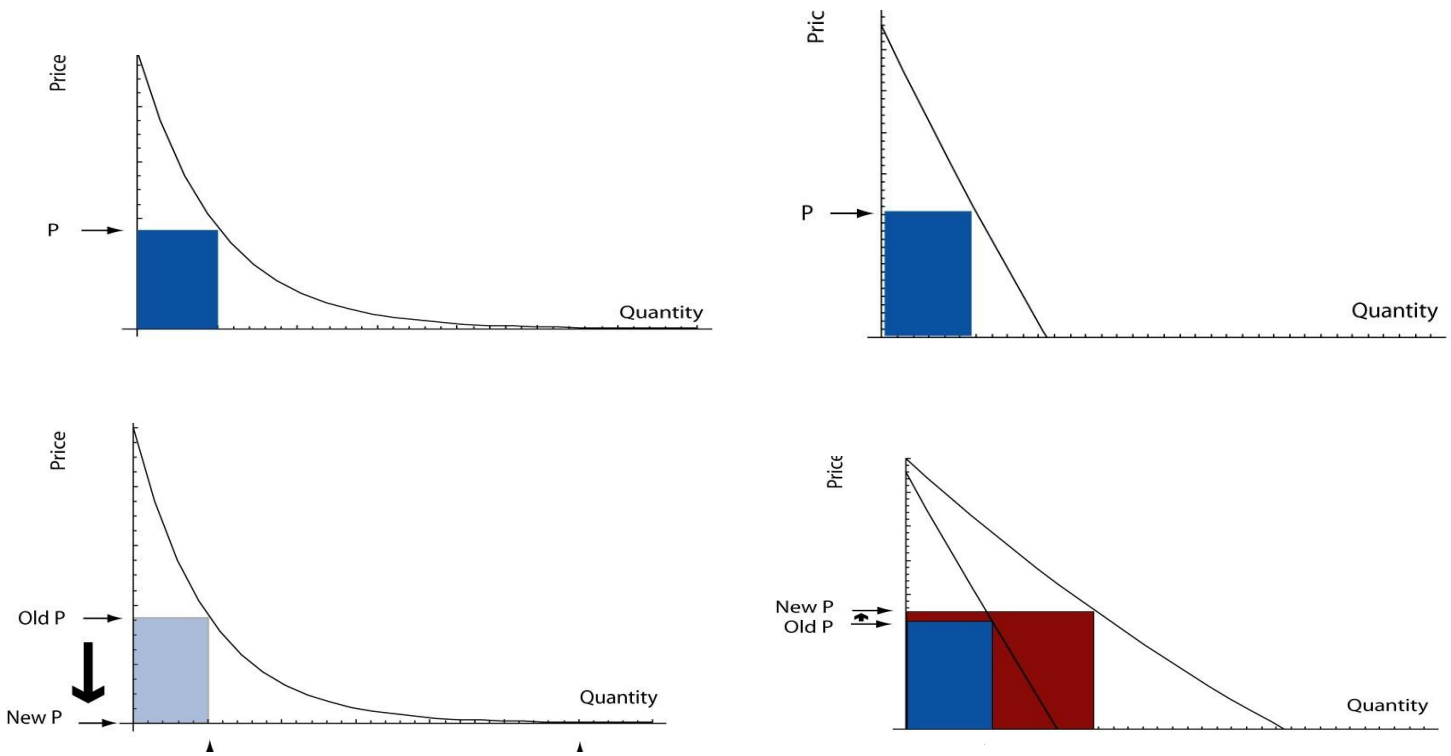
PC vs. Video Game Console: Revisit

- Different pricing strategies
  - In video games, **end users are subsidized**. Platform providers like Sony PlayStation and Microsoft Xbox price consoles at or below cost. Game developers pay a royalty to console manufacturers as much as 20% of a game retail price.
  - In the PC market, **end users are the money side**, paying well above cost. Application developers, on the other hand, pay no royalties and receive free software development kits from the OS vendors.
  - Why?

Video Game Console: Subsidize Consumers

Market One: Consumers

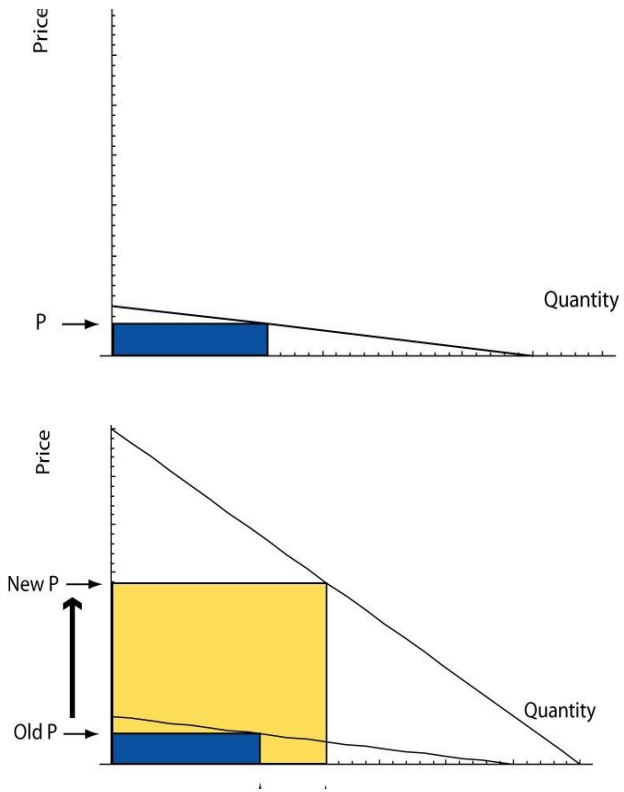
Market Two: Developers



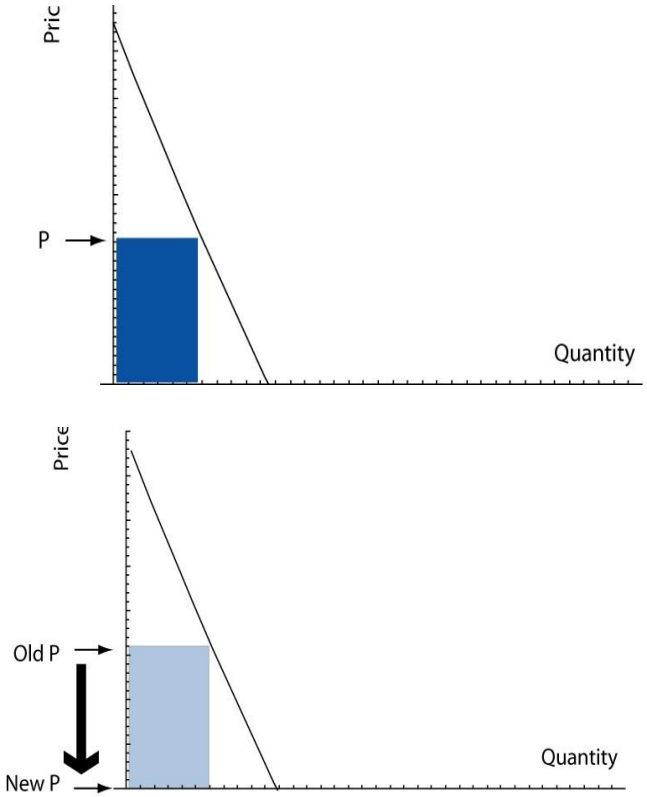
- Making game consoles more affordable dramatically increased the value of console, shifting developer demand out and to the right.
- Here, game console subsidized consumers. Profit lost on one side of market is less than profit gained on other side of market.

PC: Subsidize Developers

Market One: Consumers



Market Two: Developers



- Making System Development Toolkits (SDKs) free dramatically increased the value of operating systems, shifting consumer demand out and to the right.
- Here, Microsoft subsidized developers. Profit lost on one side of market is less than profit gained on other side.

Roadmap

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The Penguin Problem

The Launching Stage

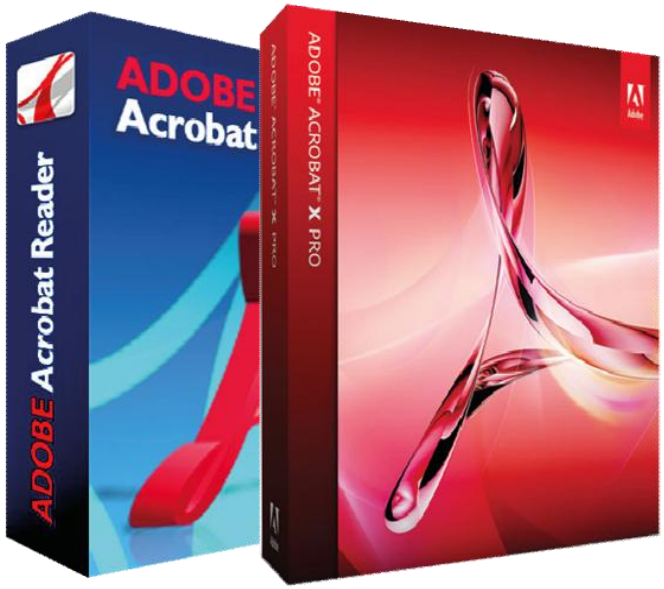
- Two-sided platforms often face a **“chicken-and-egg”** coordination problem:
  - In order to attract buyers, the platform should have a large base of sellers, and vice versa.
  - Buyer are willing to join the platform only if they expect many sellers to join the platform too.
- Problem arises when both sides hold pessimistic belief about the participation rate of the other side.
- Possible solution: **Divide-and-conquer** pricing strategy.
  - Subsidize one side of the market to get it on board and then use the presence of this subsidized side to attract the other side.

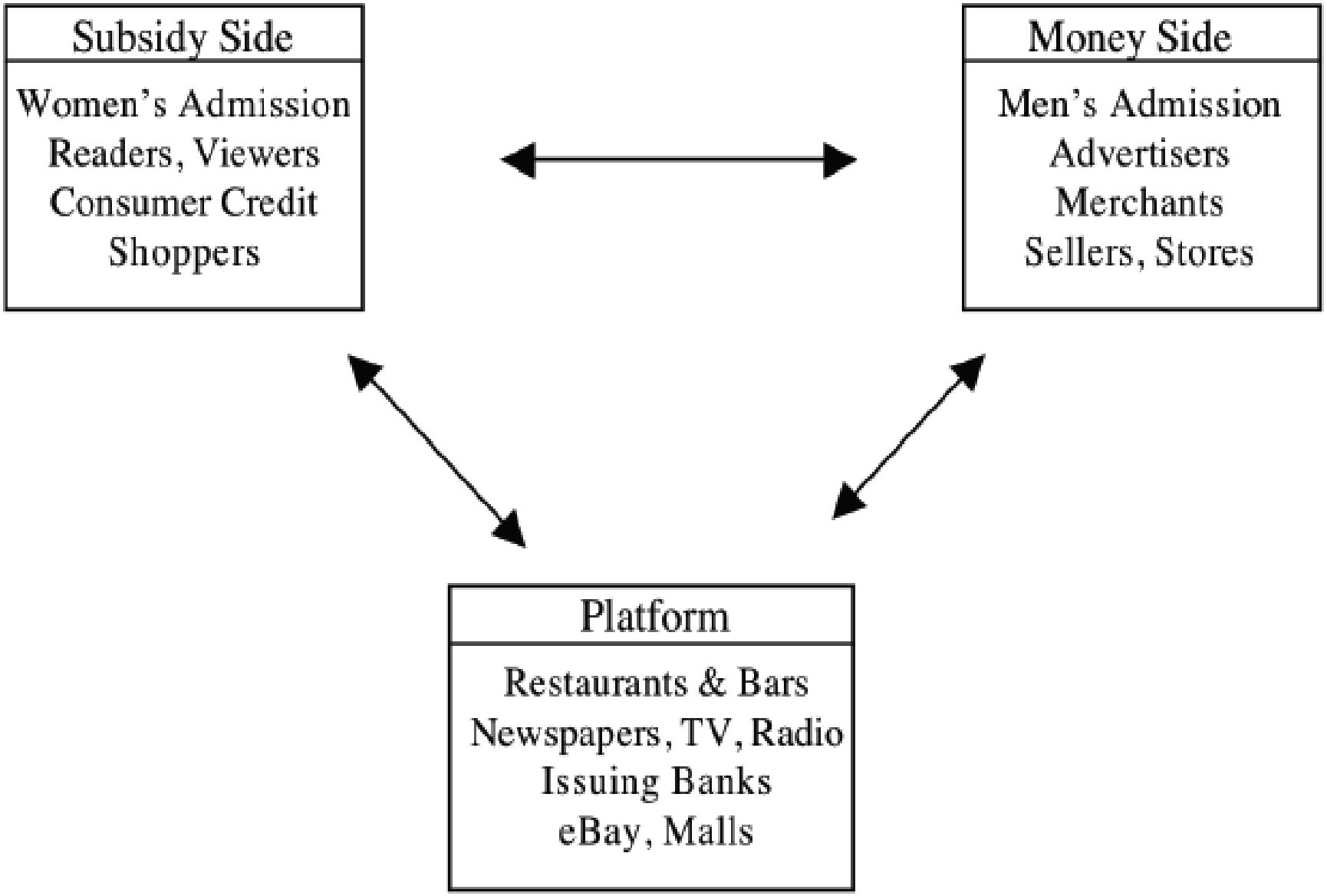
Divide-and-conquer strategy

- Equilibria with bad expectation:
  - for any $r_s > 0$ and $r_B > 0$ chosen in stage 1, sellers and buyers coordinate on the equilibrium which yields the lowest payoff to the platform.
- Divide-and-conquer strategy:
  - divide group i: $r_i \leq 0$. (Subsidy side: attract users and grow network size)
  - conquer group j: $r_j > 0, j \neq i$. (Money side: pay for the privilege of gaining access)
- How it works:
  - group i gets surplus $-r_i \geq 0$ even no group j agents join the platform. This is a free meal and they would like to join the platform.
  - given that group i agents are on board, group j agents would like to join if the r_j is reasonable. The platform then charges $r_j > 0$ to extract surplus from group j agents.

Adobe Acrobat: Revisit

- The Story of Acrobat Reader/Writer
  - Adobe Acrobat PDF reader is free while the Adobe Acrobat writer is not.
  - Why?
  - Adobe first tried to sell PDF readers and writers but no one wanted to buy readers when there was no content to read. And, no one wanted to buy writers when no one had readers. Adobe broke the logjam by giving PDF readers away.




Subsidy Side
Women's Admission
Readers, Viewers
Consumer Credit
Shoppers
Money Side
Men's Admission
Advertisers
Merchants
Sellers, Stores
Platform
Restaurants & Bars
Newspapers, TV, Radio
Issuing Banks
eBay, Malls

Divide-and-conquer Strategy

- The coordination problem is more severe for those platforms at the launching stage.
  - when a platform has established its reputation, the agents should not hold pessimistic belief about the other side's participation.

Divide-and-conquer Strategy

- In a traditional one-sided market, price depends only on **marginal cost** and **elasticity of demand**.
- But on platforms, pricing depends not only on the demand and cost from this side, but also on how their participation affects participation on the other side, and the profits associated with the other side, i.e., the elasticity and the mark-up charged to the other side.
  - Identify the group that creates large network effects for the other group and the group that is more price-sensitive

Divide-and-conquer strategy

- For a traditional one-sided market, we know that price should drop when elasticity of demand is high.
- But in a two-sided market, you want to lower it even further.
  - The increased value extracted from the other side magnifies the value of having more consumers on the price elastic side.
  - So often times, we see $p < mc$.

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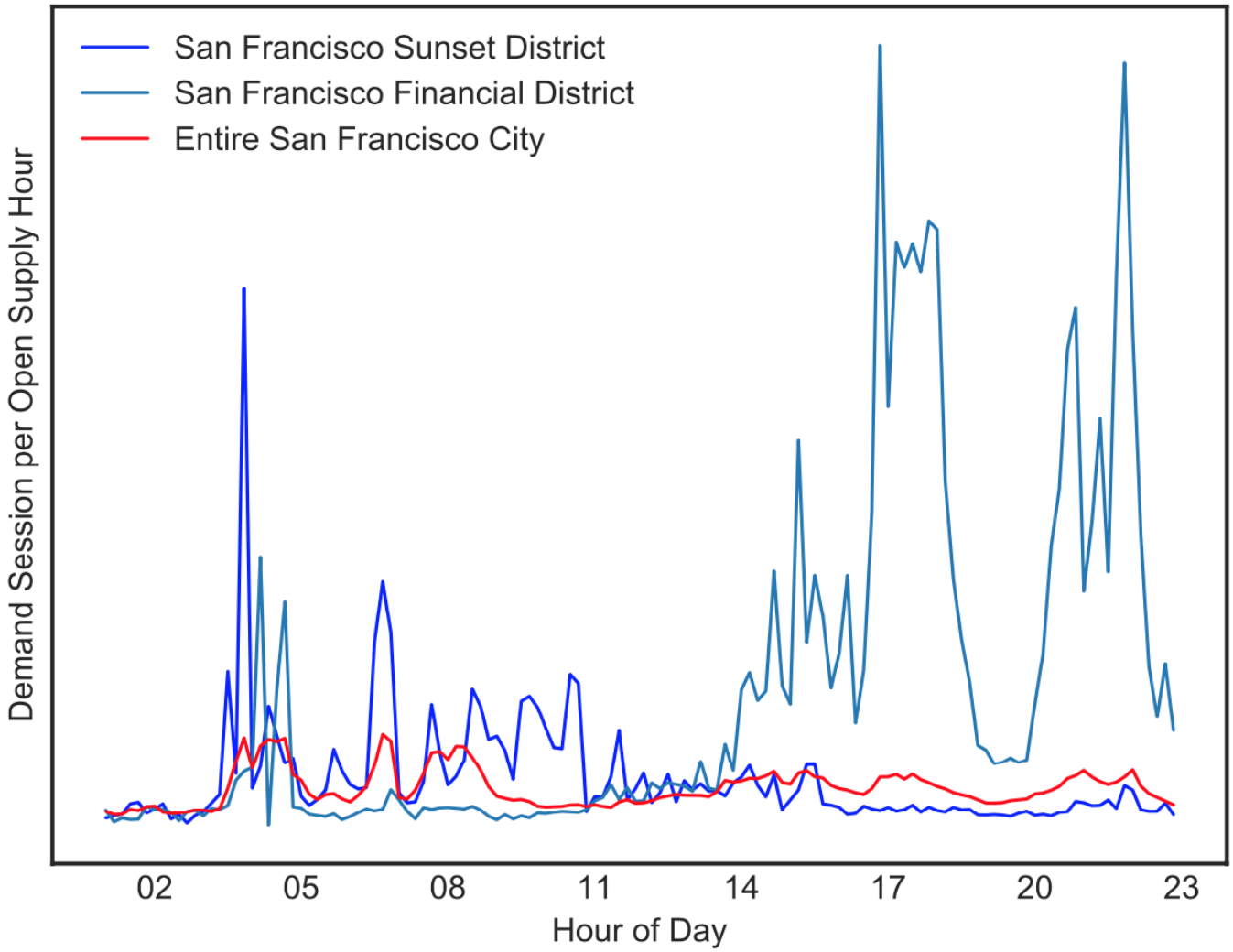
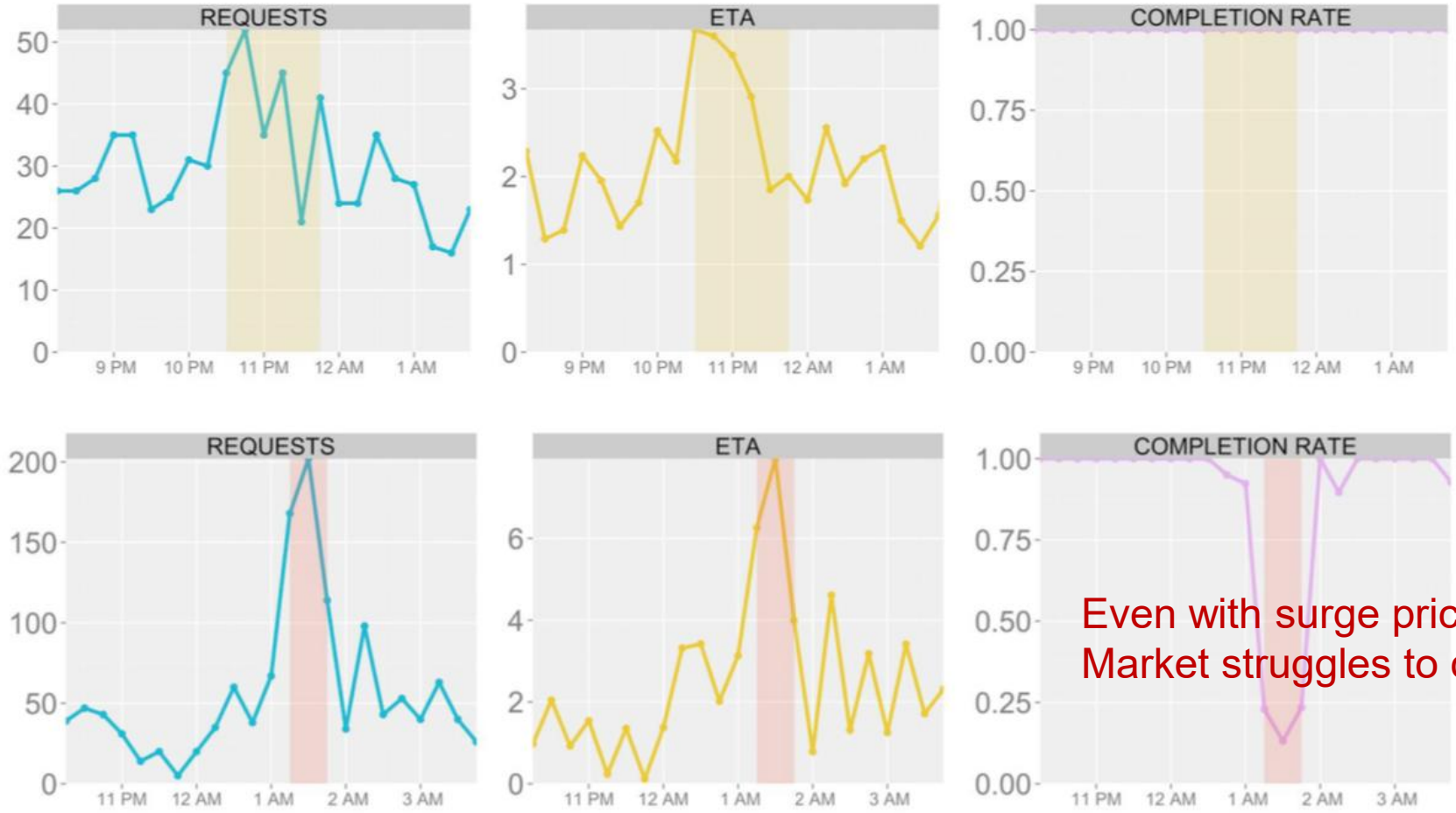
Dynamic Pricing (Surge Pricing)



Dynamic Pricing (Surge Pricing)



Even with surge pricing,
Market struggles to clear.

Surge pricing during two events, as reproduced from Hall et al. (2015). Top: after an Ariana Grande concert; Bottom: New Year's Eve 2015 in Times Square. Left: the count of requests; Middle: the en route time; Right: the fraction of requests fulfilled.

Freemium Strategy

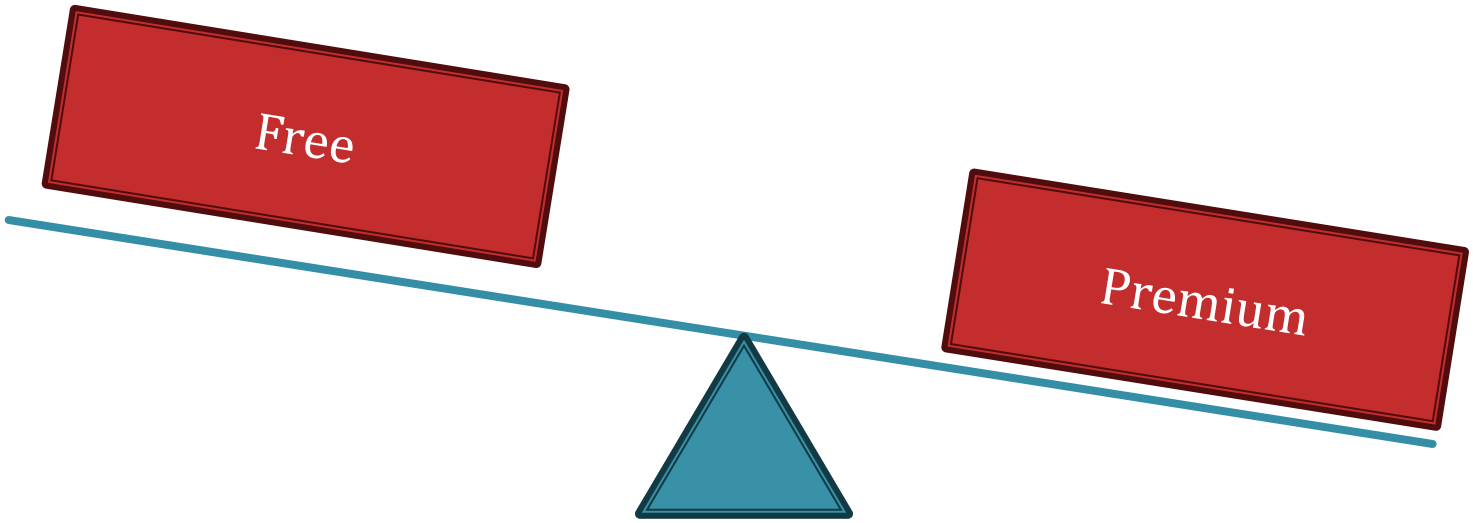
- Firms offer a mixed product strategy:
  - A premium (paid) version
  - A free version supported by advertising

Freemium Strategy

Spillover Effects and Freemium Strategy in the Mobile App Market

Yiting Deng, Anja Lambrecht, Yongdong Liu

Published Online: 30 Nov 2022 | https://doi.org/10.1287/mnsc.2022.4619



Freemium Strategy

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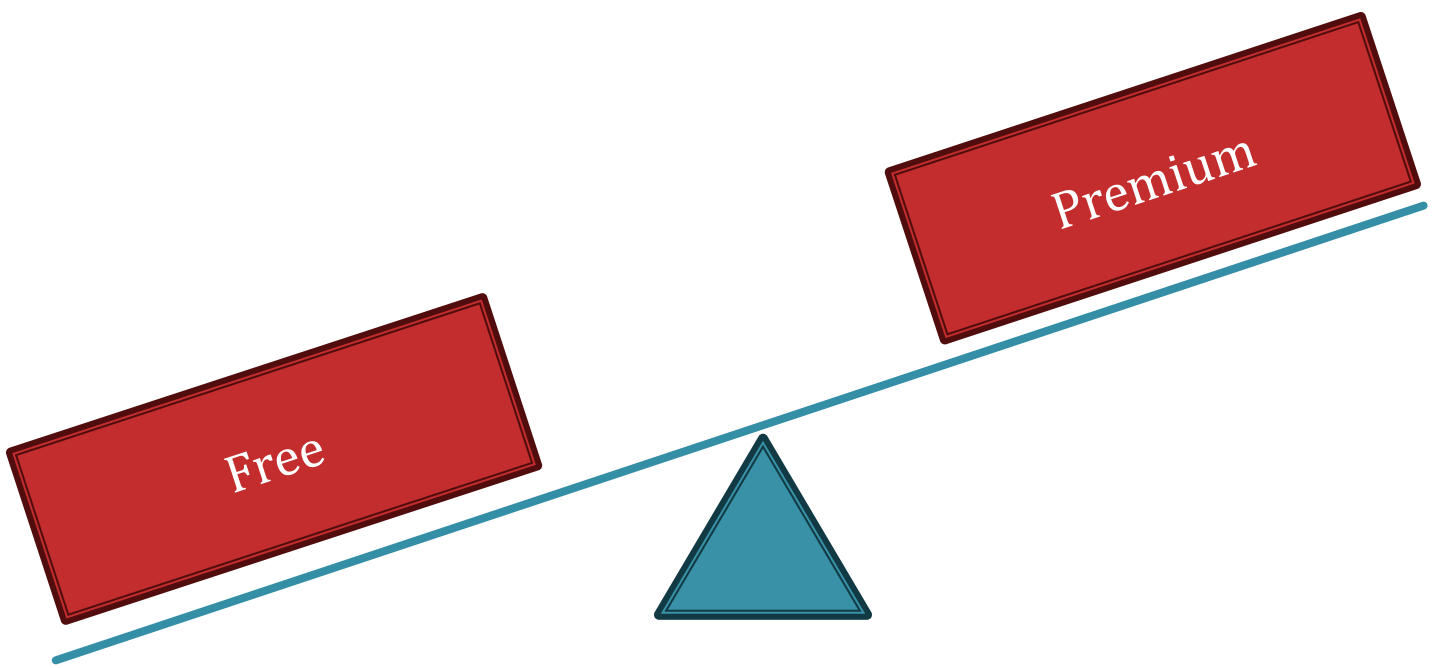


11 March Yiting's Seminar

Seminar Details

•**Topic**: **Holiday or Hangover? Housing Market Responses to a Temporary Tax Cut**

•**Speaker**: Dr. Yiting Deng

•**Date & Time**: 13:30-15:00 (UK Time), March 11, 2026

•**Venue**: Conference Room

FORTNITE

Epic Games vs. Apple

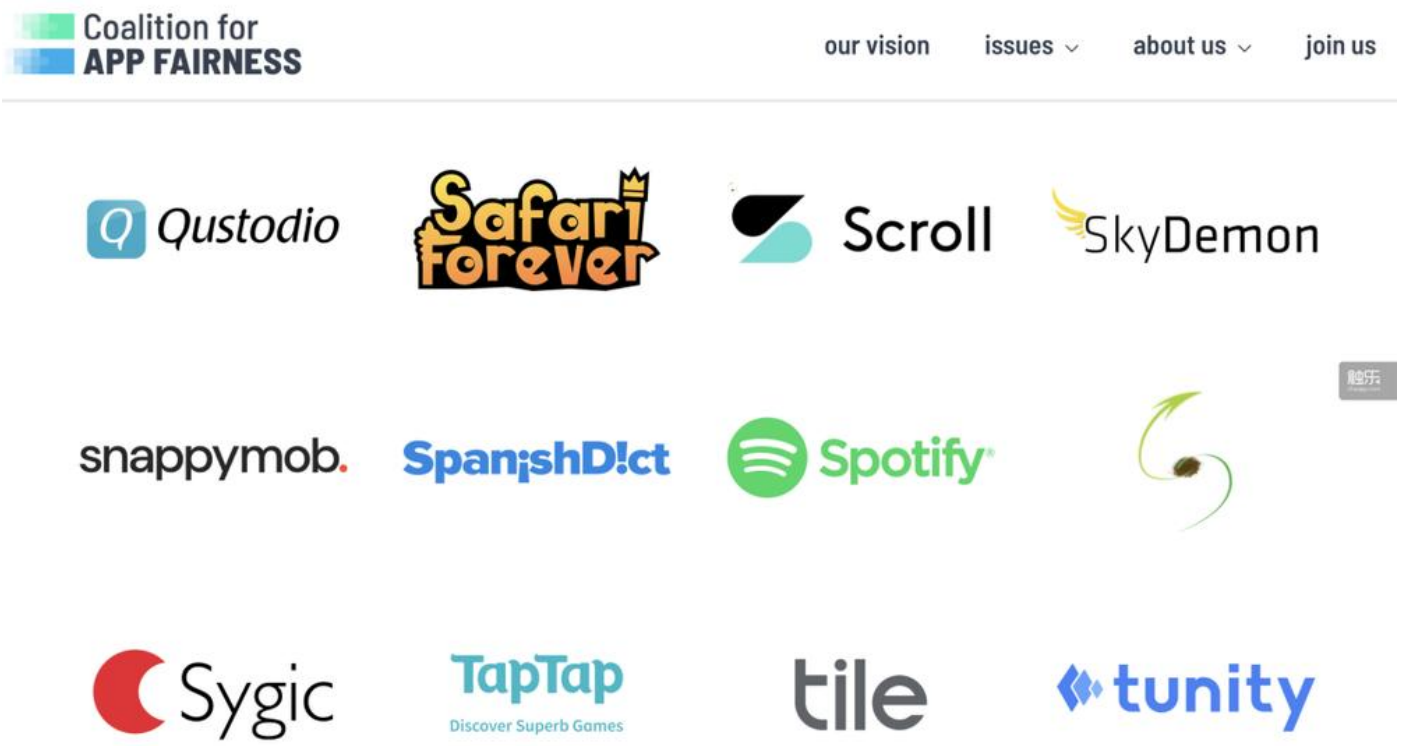
- Apple obtains 30% revenue cut from every purchase in App Store and in-game purchase
- Users of iOS and Mac can only download apps from its App Store
- Fortnite added a payment method called Epic direct payment to bypass the Apple 30% revenue cut.

Epic Games vs. Apple

- Epic Games' founder and CEO Tim Sweeney had **questioned the need for digital storefronts** like Valve's Steam, Apple's App Store for iOS devices, and Google Play, to take a 30% revenue sharing cut.
- Sweeney further stated that "The math is quite simple: we pay around 2.5 to 3.5 percent for payment processing for major payment methods, less than 1.5 percent for CDN costs and between 1 and 2 percent for variable operating and customer support costs. Fixed costs of developing and supporting the platform become negligible at a large scale. In our analysis, stores charging 30 percent are marking up their costs by **300 to 400 percent**. ”
- Apple has argued that the 30% cut it takes though In-App Purchases (IAP) "reflects the immense value of the App Store" and beyond the visible features it offers developers, that cut covers "Apple’s technology, tools, software for app development and testing, marketing efforts, platinum-level customer service, and distribution of developers’ apps and digital content.“

Epic Games vs. Apple

- Just hours later, Fortnite was taken off from App Store.
- At the same time, Epic formed a so-called Coalition For App Fairness (CAF)



Epic Games vs. Apple

Elon Musk
@elonmusk

Apple app store fees are a de facto global tax on the Internet. Epic is right.

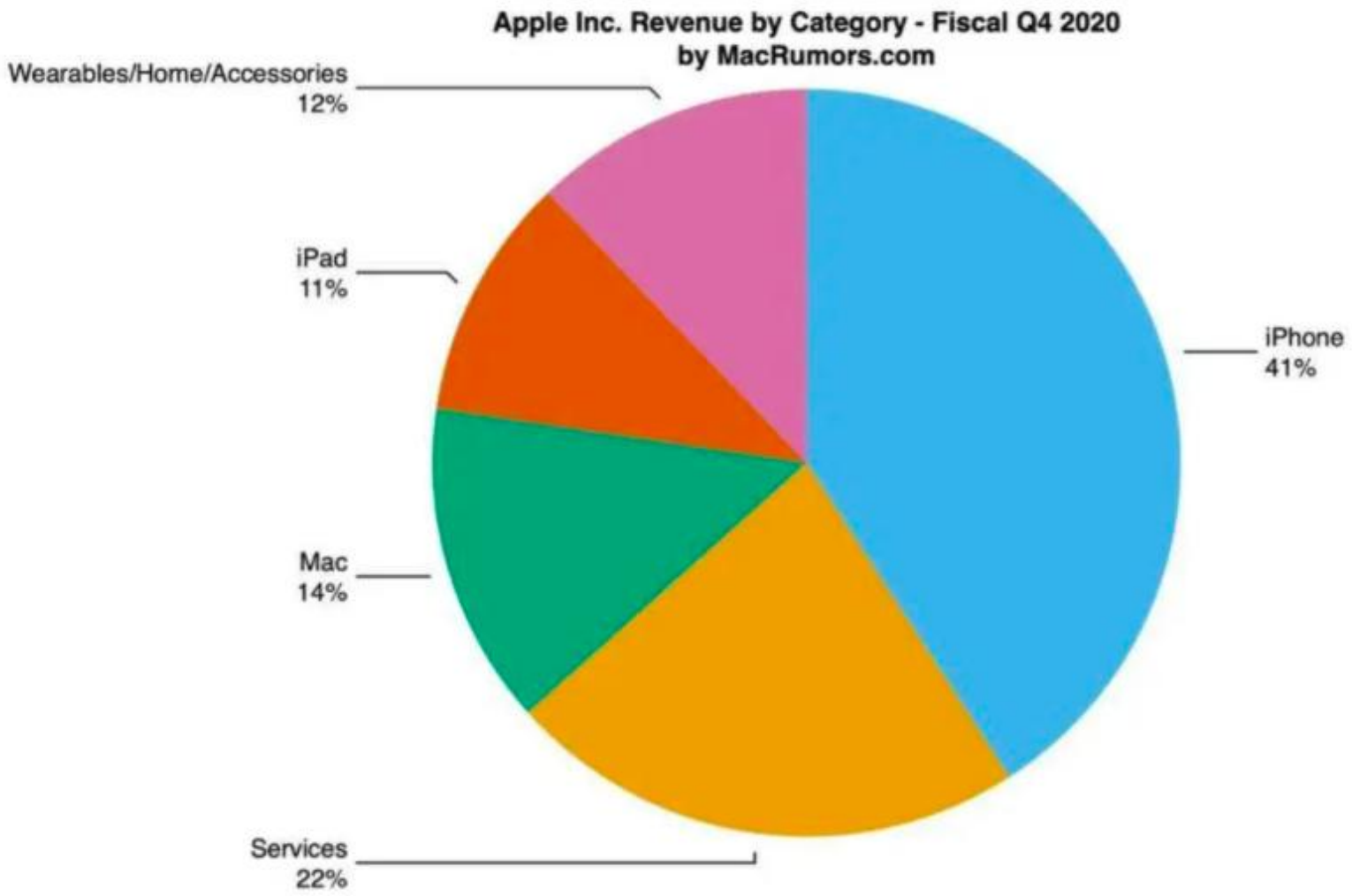
上午12:55 · 2021年7月31日 · Twitter for iPhone

1.7万 转推 2,755 引用推文 23.6万 喜欢次数

Epic Games vs. Apple

- In December 2020, Apple announced that they would be lowering the revenue cut Apple takes for app developers making $1M or less from 30% to 15% if app developers fill out an application for the lowered revenue cut

Epic Games vs. Apple



Epic Games vs. Apple

- In Sept 2021, the judge rules eventually that Apple can no longer force developers to use in-app purchasing
- The market value of Apple dropped by more than 80 billion USD immediately after the ruling.

History of 30% Revenue Cut

- In the very early time, XBOX, PSN and Nintendo did not charge for purchase at all. But later all of them started to charge fees due to profitability reason.
- 30% gradually became an equilibrium price in the game industry. Besides App Store, Google Play, Steam, GOG, store in Win10, uPlay, Origin all adopt the 30% revenue cut.
- Why 30%?

Pricing of Various Platforms

Company	Rake	Notes
OpenTable	1.9%	Reservation fee / average meal per person
Homeaway	2.5%	Estimated (low due to use of listing model instead of transaction)
Comparison Shopping	6.0%	Estimated
ebay	9.9%	This is partially listing fees, partially marketing fees, and part PayPal.
oDesk	10.0%	10% on top of all work billed
AirBNB	11.0%	3% + 6-12% depending on size of transaction
Expedia	11.9%	Per 2012 10-K
Amazon Marketplace	12.0%	Guess based on rate table
Fandango	12.5%	Fee charged to user / ticket price
PriceLine	18.5%	Per 2012 10-K
TicketMaster	26.0%	Estimate for tickets sold by TM (non box office) - very hard to discern
Steam	30.0%	Rate card
Itunes	30.0%	Rate card
Facebook Credits	30.0%	Rate card
GroupOn	38.2%	Calculated from 2012 10-K. Does not include direct goods.
Shutterstock	70.0%	From S-1

Thank you!

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